CareConnect | Database Normalization

CareConnect stores service requests, organization referrals, and follow-up records. Normalization helps us reduce repeated data and keep information consistent. The examples below use simplified records to explain the design; they are not actual user records.

1. Unnormalized Form (UNF)

Imagine storing a request and every referral attempt in one row. One cell might say: “Attempt 1: Organization 12, rejected; Attempt 2: Organization 18, accepted.” This is a repeating group. It is difficult to add or search more attempts.

2. First Normal Form (1NF)

In 1NF, each cell holds one value, and each record has a key. CareConnect stores each referral attempt as a separate row in referrals.

referral_id (PK)	request_id (FK)	attempt_number	organization_id (FK)	status
901	501	1	12	rejected
902	501	2	18	accepted

The real referrals table also has other columns. The database uses UNIQUE(request_id, attempt_number), so the same request cannot have two attempts with the same number.

3. Second Normal Form (2NF)

In 2NF, the table is already in 1NF, and a non-key value must not depend on only part of a composite candidate key.

Example: A combined table could use (organization_service_id, priority_id) as its key. Offering status depends only on organization_service_id. Priority name and deadline depend only on priority_id. Keeping them together would repeat data.

Table	Stores
organization_services	Offering details and status
priority_levels	Priority name and deadline_hours
service_priority_support	Which offering supports which priority (is_supported)

The pair (organization_service_id, priority_id) is unique in service_priority_support. The support value belongs to that full pair. A separate support_id is its primary key.

4. Third Normal Form (3NF)

In simple terms, 3NF avoids storing facts about another non-key field in the same table. A formal 3NF check also considers all candidate keys and functional dependencies.

Example: If users stored role_id, role_name, and role description, the role details would repeat for many users. Instead, users stores role_id (FK), while roles stores role_name and description. We update a role description in one place.

Main table	Related table	Information kept separately
users	roles	Role name and description
services	service_categories	Category name and description
service_priority_support	priority_levels	Priority name and deadline

5. Functional Dependencies

A functional dependency means one value determines another value. These are examples from the database:

Dependency	Meaning
role_id → role_name, description	One role ID identifies one role.
priority_id → level_name, deadline_hours	One priority ID identifies its rules.
referral_id → request_id, attempt_number, status	One referral ID identifies one referral.
(request_id, attempt_number) → referral_id	Each attempt number is unique within a request.

6. Conclusion

The schema has 20 domain tables. The newer version described in the team document also includes event_outbox for RabbitMQ events, giving 21 tables in total. Confirm this migration is present in the database being demonstrated.

Some values are stored for workflow needs. For example, is_backup follows the attempt number, and active_slot is generated from referral status in the newer migration. These need clear rules and should not be used to claim that every table is strictly proven to be in 3NF.

Conclusion: CareConnect uses 1NF, 2NF, and 3NF design principles to reduce repeated data. Related information is stored in separate tables and connected using foreign keys. A complete 3NF claim would need a full check of the business rules and functional dependencies.

Suggested speaking time: 2–3 minutes. Focus on one example for each normal form; use the remaining time for the ER diagram and SQL queries.